An Execution-Time CBF-QP Safety Filter on Frozen MAPPO Formation Policies: A Pre-Specified Paired Evaluation in Simulation

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DRAFT - not for distribution; authorship and framing pending advisor review

AI-assistance statement. The code, analyses and this draft were produced with substantial assistance from an AI coding assistant (Claude, Anthropic) under the author's direction. Every number in the text is generated by committed scripts from committed result files (`scripts/paper_numbers.py`, `paper/numbers.tex`). The author is responsible for the content.

Abstract

We evaluate whether an execution-time, decentralised control-barrier-function quadratic-programme (CBF-QP) safety filter, placed between frozen multi-agent PPO (MAPPO) formation policies and a simulated differential-drive team, is useful under criteria fixed before the test. Nine frozen actors (three training seeds, three checkpoints each) were evaluated on reserved, previously unused layouts with one, two and three obstacles (9,000 paired trials per density), with the filter off, the filter on at the gain $\alpha^\star = 10$ selected on validation layouts, and a brake comparator. The filter reduced the share of trials ending in collision or infeasibility at every density (from 5.4% to 0.1%, 16.6% to 0.8% and 48.0% to 3.9%). Under the pre-set criteria it was nevertheless *not shown useful* at any density: the non-inferiority criteria for goal completion (margin five percentage points) and for formation error over the common executed prefix (margin 0.01 m) were not met. Exploratory, post hoc analyses of the saved logs indicate that many filtered runs stalled in front of obstacles on layouts the unfiltered policy completed or collided on, and that in all 433 infeasible trials the binding barrier row sat at its stand-off, consistent with numerical sensitivity at $\alpha\Delta = 1$. Claims cover these nine actors in this simulated scenario only.

1 Introduction

Reinforcement-learning policies for multi-robot formations are commonly trained without safety constraints, and a safety filter can be added at execution time without retraining. Whether such a filter is *useful*, i.e. reduces failures without an unacceptable loss of task completion or formation quality, is an empirical question for a given policy family. This draft reports a pre-specified paired evaluation of one such filter, a decentralised sampled-data CBF-QP, on frozen MAPPO actors in a VMAS differential-drive formation-navigation scenario, followed by clearly labelled exploratory analyses of why the pre-set criteria were not met.

2 Related work

Execution-time filtering of policies trained without the filter appears in [1] (centralised QP, linear and double-integrator agents); filtering during training and execution with model-predictive shields for unicycle formations appears in [2]. Sampled-data CBF margins that account for inter-sample behaviour are studied in [3, 4, 5, 6]; these margins are input-independent bounds over the input set and do not depend on the applied command. Decentralised equal-split ("half-responsibility") constraints go back to [7, 8], and stopping as a feasibility-preserving backup is used in [9]. The offset (barrier) point for unicycles is used in [10]. Hidden assumptions in CBF safety claims are discussed in [11]. Input-dependent inter-sample bounds used for trigger timing appear in [12].

3 System and filter

Three differential-drive robots (radius 0.10 m, $v \in [-0.5, 0.5]$ m/s, $|\omega| \le 1$ rad/s) move a triangular formation (slot radius 0.30 m) through a field of circular obstacles (radius 0.20 m) to a goal; control period $\Delta = 0.1$ s, horizon 400 periods. GOAL requires the centroid within 0.10 m of the goal and mean slot error within 0.10 m; COLLISION is scored on the swept path and ends the trial; an infeasible QP ends the trial as INFEASIBLE before any motion; otherwise TIMEOUT. The filter ("filter C") constrains a point offset $\ell = 0.05$ m ahead of each axle with barriers $h = \|p - c\|^2 - (R + \ell + R_o)^2$ (obstacles) and $h = \|p_i - p_j\|^2 - (2(R + \ell))^2$ (pairs, equal split), a sampled-data margin from a two-sided bound on $\ddot{h}$ (design document, Lemma 1), and solves one two-variable QP per robot exactly. The B-match comparator applies $(0, \omega_{\text{nom}})$ whenever the clipped nominal command violates one of the filter's rows at the same gain; it is not a certified stop (the offset point rotates).

4 Experimental design

The protocol was locked by the author before any filter-on data existed (the margins and decisions were chosen without advisor review; see Section 9). Nine actors were fixed by a checkpoint rule (early/middle/late of the final curriculum stage, seeds 0–2). The gain was selected on existing validation layouts from $\alpha \in \{0.5, 1, 2, 5, 10\}\,\text{s}^{-1}$ by the highest pooled filter-on GOAL count over the nine actors and $K = 1, 2, 3$. The final test used 1,000 reserved layouts per actor and density, each run with the filter off, the filter on at $\alpha^\star$ and B-match at $\alpha^\star$. Usefulness at a density required all of: (a) a one-sided exact McNemar reduction in COLLISION-or-INFEASIBLE; (b) GOAL loss (off minus on) with one-sided 98.33% upper bound (Newcombe method 10) at most 0.05; (c) the equal-actor-weighted mean increase in formation error over the common executed prefix with stratified bootstrap 98.33% upper bound at most 0.01 m; each one-sided at 0.05/3.

5 Results (confirmatory)

Gain selection (development data). Pooled filter-on GOAL counts over 5,400 validation trials were 0, 255, 1,184, 1,936 and 2,224 for $\alpha = 0.5, 1, 2, 5, 10$, with 1,251, 4, 11, 61 and 129 INFEASIBLE outcomes, so $\alpha^\star = 10$ (Fig. 1).

Table 1 gives the confirmatory results. Criterion (a) was met at every density; (b) and (c) were met at none, so the filter was *not shown useful under the pre-set criteria* at $K = 1, 2, 3$. With the filter on there were 0, 0 and 0 collisions and 6, 75 and 352 INFEASIBLE trials; TIMEOUT rose from 39.9%, 16.9% and 20.0% to 50.1%, 42.4% and 74.6% (Fig. 2). B-match had 0, 0 and 0 collisions and 0, 0 and 0 INFEASIBLE outcomes, and reached GOAL in 48.8%, 53.5% and 19.3% of trials; the

Table 1: Confirmatory results, filter on ($\alpha^\star = 10$) versus filter off, nine actors pooled, 9,000 pairs per density. Differences in percentage points (pp); one-sided bounds at 98.33%; 95% intervals descriptive.

	$K = 1$	$K = 2$	$K = 3$
COLLISION-or-INFEASIBLE, off → on (%)	5.4 → 0.1	16.6 → 0.8	48.0 → 3.9
(a) difference on−off [pp]	−5.30	−15.72	−44.08
98.33% upper; 95% CI	−4.82; [−5.78, −4.85]	−14.92; [−16.49, −14.98]	−42.95; [−45.11, −43.04]
one-sided exact p; criterion	2.6×10^{-144}; met	$< 10^{-300}$; met	$< 10^{-300}$; met
GOAL off / on (%)	54.7 / 49.9	66.6 / 56.8	32.0 / 21.5
(b) GOAL loss off−on [pp]	4.81	9.81	10.56
98.33% upper (margin 5); criterion	5.31; not met	10.59; not met	11.33; not met
(c) formation increase [m]	0.0117	0.0202	0.0278
bootstrap 98.33% upper (margin 0.01)	0.0127	0.0213	0.0291
normal cross-check; eligible / excluded	0.0126; 9,000/ 0	0.0213; 9,000/ 0	0.0291; 9,000/ 0
criterion	not met	not met	not met
Verdict	not shown useful	not shown useful	not shown useful

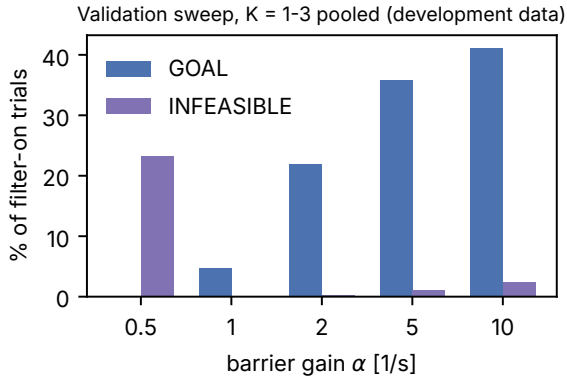


Figure 1: Validation sweep (development data): filter-on GOAL and INFEASIBLE share per gain, $K = 1$–3 pooled.

filter reached GOAL 1.0, 3.2 and 2.2 percentage points more often (descriptive). Per-actor GOAL losses vary widely (Fig. 3); a pooled result does not establish usefulness, or its absence, for each actor.

6 Exploratory mechanism analysis (exploratory, post hoc)

Everything in this section is exploratory and post hoc: categories, thresholds and groups were chosen after the confirmatory results were known, the analyses read only saved logs, and they support no causal or confirmatory claim.

Stall classes. Of 15,038 filter-on TIMEOUTs, 9,211 (61%) were classified as filter-induced stalls (the clipped nominal command violated an enabled barrier row in at least half of the last 50 periods), 9,143 of them involving obstacle rows; on the same layouts the unfiltered run collided in 5,699 and reached GOAL in 2,274 cases. Another 5,670 (38%) were near-goal parking stalls, 5,362 of them on layouts where the unfiltered run also timed out.

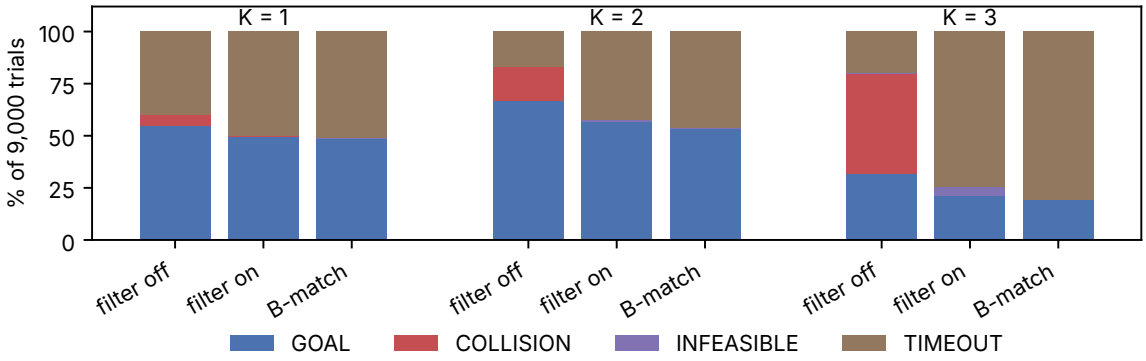


Figure 2: Final-test outcome shares per density for filter off, filter on ($\alpha^\star$) and B-match.

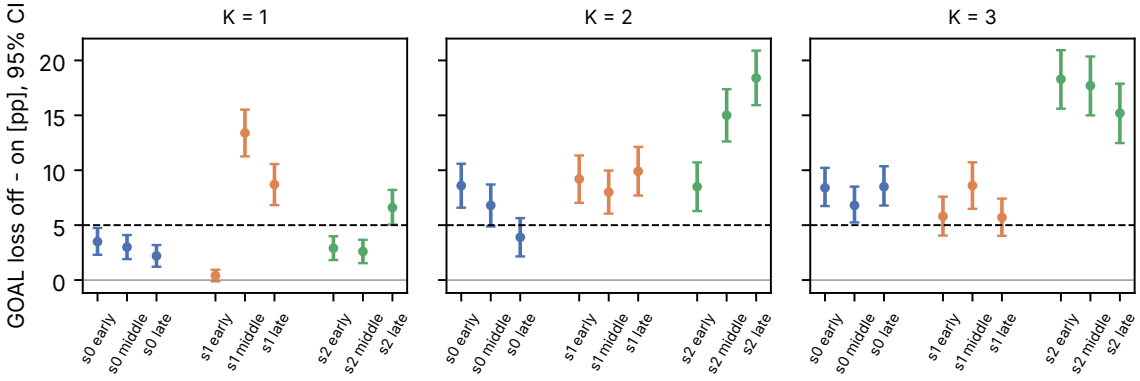


Figure 3: Per-actor GOAL loss (off minus on) with descriptive 95% Newcombe intervals, grouped by training seed; dashed line: the 5 pp margin of the pooled criterion (b).

Groups A and B. Group A (2,274 pairs: unfiltered GOAL, filtered filter-induced stall; 2,259 obstacle-labelled, 15 neighbour-only) and group B (11,255 pairs: both GOAL) were selected by observed outcome. The unfiltered runs of group A came closer to obstacles (median sampled minimum surface clearance 3.3 cm, group A only) than those of group B (14.4 cm; Fig. 4). Their applied commands violated at least one filter obstacle row in 89% (A) versus 4% (B) of runs, and a robot-pair row in 85% versus 36%.

Clearance thresholds. Of all trials, 51.1% reached GOAL unfiltered, and 43.1% did so while keeping a sampled minimum obstacle clearance of at least 5 cm, against a filter-on GOAL rate of 42.7% on the same layouts (Fig. 4). This describes how close the unfiltered runs came to obstacles; it does not measure what a buffer or any filter would cost. Among unfiltered GOAL trials the descriptive AUC of the unfiltered minimum obstacle clearance for "filtered run did not reach GOAL" was 0.96.

Infeasibility. All 433 INFEASIBLE trials were reproduced (433 of 433) by the unchanged filter code at the saved states. In every case the binding row's barrier value was within $4.3 \times 10^{-8}\,\mathrm{m}^2$ of its stand-off $h^\star = \delta/\alpha$ and the zero command violated it by between -8.4×10^{-9} and -4.9×10^{-6}

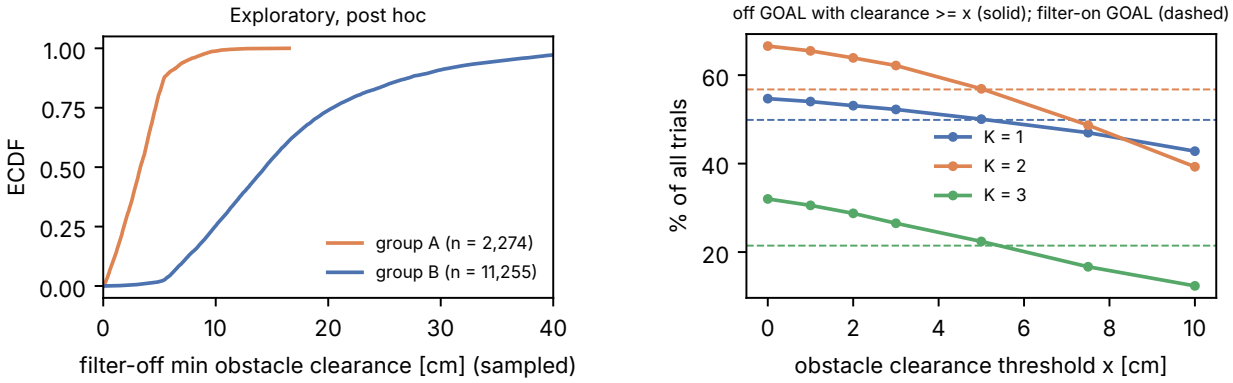


Figure 4: Exploratory, post hoc. Left (F4): unfiltered minimum obstacle clearance (sampled), group A vs group B. Right (F5): share of all trials that reached GOAL unfiltered with clearance at least x, per density, with the filter-on GOAL rate (dashed).

(normalised); 410 of 433 met a pre-declared near-boundary diagnostic. This is consistent with numerical sensitivity near the boundary, not proof of the cause.

Transit. During formation transit the equal-split cap on a robot's closing speed toward a neighbour had median 0.72–0.78 m/s, above the largest barrier-point speed 0.50 m/s, and bound in 1–8% of moving closing periods.

7 Discussion

Collisions versus stalls. With the filter on there were no collisions (0, 0 and 0 at $K = 1, 2, 3$; the remaining failures were INFEASIBLE; Table 1); in the exploratory classification most filtered timeouts were stalls with active obstacle rows, mostly on layouts where the unfiltered run collided or reached GOAL (stall classes above). We read this as a description of where the lost completions occurred, not as a causal attribution.

Small clearances. The completions lost in group A involved unfiltered runs that passed close to obstacles (median 3.3 cm, group A only; Fig. 4).

Comparator. The filter reached GOAL 1.0–3.2 percentage points more often than B-match on the same layouts, with more INFEASIBLE outcomes (results above).

Gain at the lemma boundary. The selected gain gives $\alpha^\star\Delta = 1$. At $\alpha^\star\Delta = 1$, the lemma provides no strictly positive endpoint lower bound from the initial barrier alone. Numerical robustness requires separate analysis. The infeasibility audit found the recorded infeasible states consistent with numerical sensitivity near this boundary (infeasibility paragraph).

Predictions. A methodology report written before the validation sweep, but committed to the repository only after the final test (`docs/methodology/README.md`), predicted that the selected gain would lie in $[5, 10]$ and that small gains would conflict with holding the formation; the sweep selected $\alpha^\star = 10$ and gave 0 filter-on GOAL outcomes with 1,251 INFEASIBLE at $\alpha = 0.5$ (Fig. 1). Because of the commit order these are documented, not pre-registered, predictions.

Remark (ZR). For these offset-point barriers the second-order sampled-data margin can be evaluated at the held command and written as two affine rows per barrier, which makes the zero command admissible whenever $h \geq 0$ in exact arithmetic (design document, Section 4a). Its

ingredients are existing ideas [4, 8, 9], the inclusion of admissible sets is non-strict, and the property does not survive float32 rounding without a separate tolerance rule; it was not implemented or evaluated here.

8 Limitations

The study uses one simulated scenario, nine fixed actors from one custom MAPPO implementation, three obstacle densities and one gain. Validation layouts were reused for gain tuning. The formation criterion covers the common executed prefix only. Safety statements are numerical under the design-document assumptions, not a continuous-time proof. The exploratory analyses are selected by observed outcome and descriptive.

9 Mandatory disclosures

- The margins and protocol decisions were chosen by the author without advisor review; advisor agreement has not been obtained.
- Filter-off validation results had been seen before the protocol was locked.
- The checkpoint rule was committed before the Phase 1 evaluation but after earlier validation results were seen.
- The validation layouts were reused as development data for gain tuning.
- The methodology report was committed after the final test.
- The mechanism categories were chosen after the confirmatory results were known.
- The post-test decision rule was not committed before the mechanism results were read.
- One of the 27,000 final-test trials (s2_early, $K = 3$, trial 646) started with a negative obstacle cover barrier ($h = -0.00777\,\mathrm{m}^2$, body clearance $0.087\,\mathrm{m}$), an exception to assumption A5; it is kept in all confirmatory numbers.
- Claims cover these nine fixed actors in this simulated scenario only.
- Safety statements are numerical under the design-document assumptions (A1–A9), not a continuous-time proof.
- The MAPPO implementation is custom and was sanity-checked only on the stock VMAS navigation task.

10 Conclusion

Under the pre-set criteria, the execution-time CBF-QP filter at $\alpha^\star = 10$ was not shown useful for these nine frozen MAPPO actors at any of the three densities: it reduced collision-or-infeasible failures but did not meet the non-inferiority criteria for goal completion and common-prefix formation error.

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